STATS 413: LECTURE 12

1. Influential observations
2. Leverage scores
3. Testing for outliers
4. Cook's distance

RECAP: RESIDUALS AND THE STRONGER LINEAR MODEL

Suppose the stronger linear model holds,

$$y = X\beta + \varepsilon; \quad \varepsilon \sim \mathrm{MVN}(0, \sigma_\varepsilon^2 I_{n\times n}),$$

- What's $\mathbf{E}(e_i \mid x_i) = \mathbf{E}(y_i - x_i^\top \hat{\beta} \mid x_i)$?
- What's $\mathbf{var}(e_i \mid x_i)$?
- What was the distribution of e_i?
- Is e_i independent of e_j? Are $e_1, \ldots, e_n$ identically distributed?
- What were standardized residuals, r_i?

RECAP: DIAGNOSTICS FOR THE STRONGER LINEAR MODEL

The stronger linear model has the following core components:

1. **Linearity**: At any $x_i, \mathbf{E}(y \mid x_i) = \beta_0 + \beta_1 x_{i1} + \ldots + \beta_p x_{ip} = x_i^\top \beta$
2. **Homoskedasticity**: For any $x_i, \mathbf{var}(\varepsilon_i \mid x_i) = \sigma_\varepsilon^2$ for all i.
3. **Normality**: ε_i are IID normally distributed

RECAP: DIAGNOSTICS FOR THE STRONGER LINEAR MODEL

We discussed the use of diagnostic plots to assess each of these assumptions.

- What plot(s) do we use to assess linearity? What are we hoping to see in these plots?
- What plot(s) do we use to assess homoskedasticity? What are we hoping to see in these plots?
- What plot(s) do we use to assess normality? What are we hoping to see in these plots?

RECAP: IMPORTANCE OF THE ASSUMPTIONS

Consider our test statistic for testing the null $H_0 : \beta_j = c$ with a two-sided alternative

$$t_{stat} = \frac{\hat{\beta}_j - c}{\text{se}(\hat{\beta}_j)}$$
$$p_{val} = P(|T| \geq |t_{stat}|),$$

where T follows a t distribution with $n - p - 1$ degrees of freedom, and $\text{se}(\hat{\beta}_j) = \hat{\sigma}_\varepsilon \{(X^\top X)^{-1}_{(j+1),(j+1)}\}^{1/2}$

- Where does linearity play a role?
- Where does homoskedasticity play a role?
- Where does normality play a role?

TODAY: FINDING UNUSUAL DATA POINTS

In addition to our suite of regression diagnostics assessing linearity, homoskedasticity, and normality, we'll introduce additional diagnostics to help us flag potentially unusual / important observations in our data set

- **Outliers** - observations do not behave like the bulk of our data points
- Seem to represent a departure from the trends observed in the majority of our observations.

TODAY: FINDING UNUSUAL DATA POINTS

- **Influential points** - observations which unduly/substantially affect the prediction equation.
- Not all observations have the same impact on the OLS solution $\hat{\beta}$.
- What makes a point matter more or less?
  A data point could be none, one, or both of these.

OUTLIERS

Informally, an **outlier** is an observation that differs substantially from the other points in the data set.

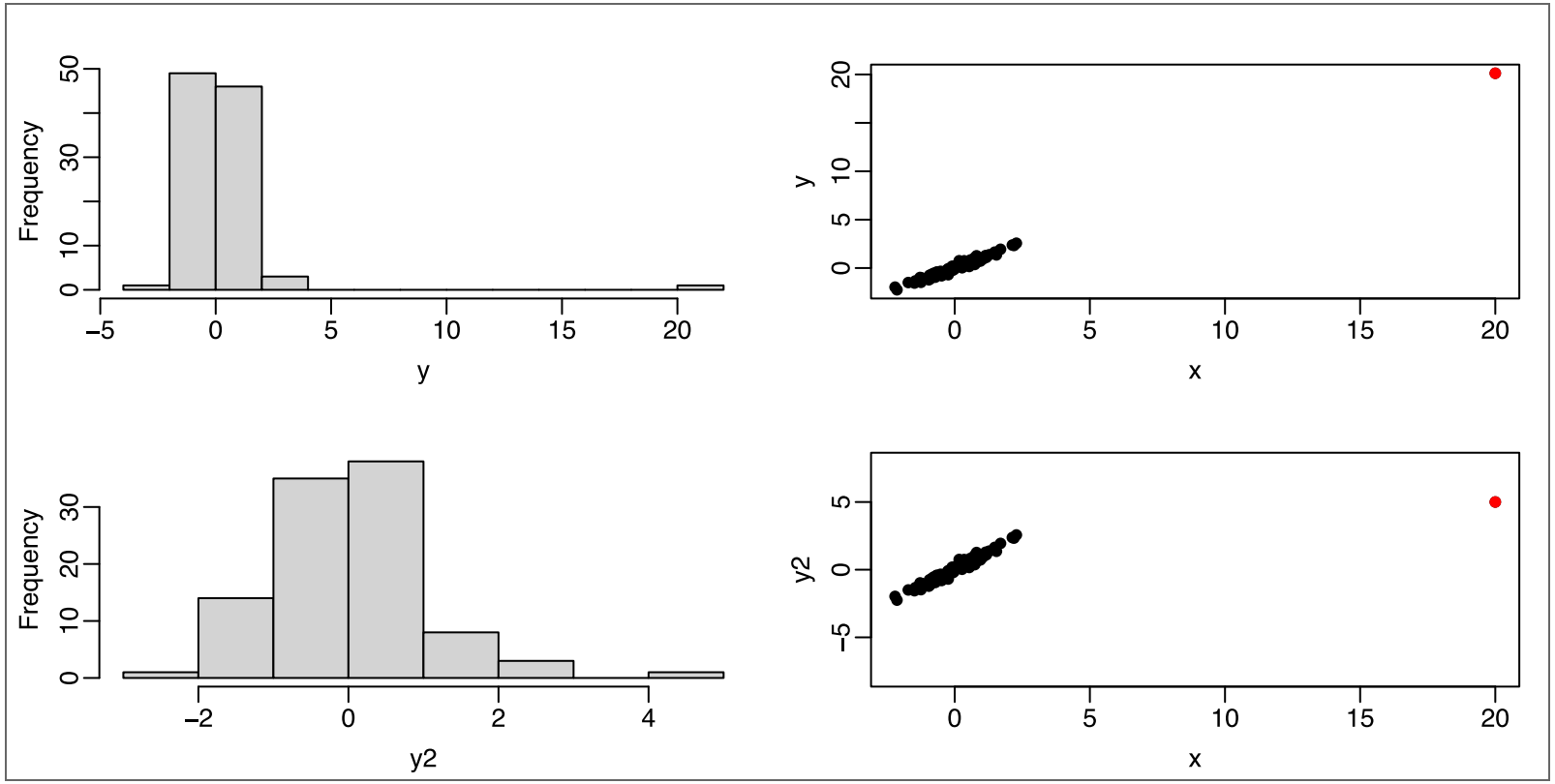
- Might be due to misrecording or typo when inputing the data, or due to a failed experiment
- Might also represent a rare, but possible, outcome in our population of interest (might be an exciting discovery!)
- Could represent that something has gone awry (for instance, credit card fraud in the distribution of one's monthly expenditures).

OUTLIERS

Informally, an **outlier** is an observation that differs substantially from the other points in the data set.

- Might be due to lurking variables (for instance, if a professional athlete is included in an investigation of exercise habits).
- Including an additional predictor variable can also **reveal** outliers that weren't apparent before.

POTENTIAL OUTLIERS AND LURKING VARIABLES

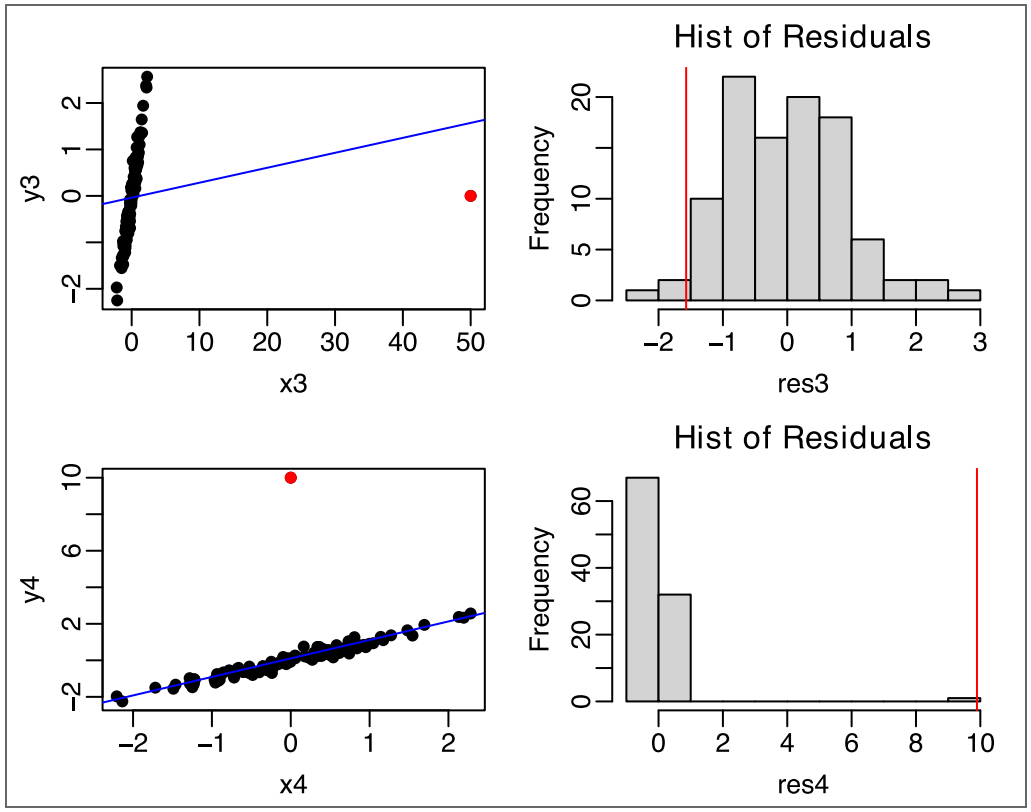


FLAGGING OUTLIERS NUMERICALLY

When conducting a linear regression analysis, can we think of a numerical way to flag potential outliers?
First thought: look for observations with extremely large or small residuals, $e_i = y_i - x_i^\top \hat{\beta}$

- Hope: individuals who deviate substantially from the majority of the data should have residuals that "stand out"
- We could simply assess the distribution of the residuals, look for observations which are extreme.
- Does this strategy work?

RESIDUALS AND OUTLIERS



The blue lines are the least squares estimates of the line of best fit for these two data sets...

SELF-INFLUENCE / LEVERAGE

Individual observations vary in the influence they have over the fitted values, and over the estimated regression line.

- From the previous slide, we see evidence that whether a not a data point's predictor value is "extreme" plays a key role here
  We define the **leverage**, or **self-influence**, of the ith observation (with covariate x_i) as

$$\text{Leverage}_i = h_{i,i},$$

where $h_{i,i}$ is the ith diagonal entry of the hat matrix,

$$H = X(X^\top X)^{-1}X^\top.$$

SELF-INFLUENCE / LEVERAGE

$$h_{i,i} = \boldsymbol{x}_i^\top (X^\top X)^{-1} \boldsymbol{x}_i$$

- Leverage thus depends not only on $\boldsymbol{x}_i$, but on the predictor variables for the other $n - 1$ individuals in the data set.

LEVERAGE AND FITTED VALUES

$h_{i,i}$ serves as a measure of how much the ith individual's observed value, y_i, influences the ith observation's fitted value, $\hat{y}_i$.

$$\begin{aligned}\hat{y} &= Hy \\ \hat{y}_i &= \sum_{j=1}^{n} h_{i,j} y_j \\ &= h_{i,i} y_i + \sum_{j \neq i} h_{i,j} y_j\end{aligned}$$

When conducting a regression with an intercept, we have the following properties

- $1/n \leq h_{i,i} \leq 1$
- $\sum_{i=1}^{n} h_{i,i} = (p+1)$
- $h_{i,i}(1 - h_{i,i}) = \sum_{i \neq j} h_{i,j}^2$ (comes from idempotence, $HH = H$).

INTUITION FOR LEVERAGE

Fitted value:

$$\hat{y}_i = h_{i,i} y_i + \sum_{j \neq i} h_{i,j} y_j$$

$h_{i,i}$ can be thought of as the weight placed on y_i when forming $\hat{y}_i$. $\sum_{j \neq i} h_{i,j} y_j$ is the contribution from the other $n - 1$ observations.

Consider the extreme case $h_{i,i} = 1$:

- What happens to $\hat{y}_i$?
- What happens to e_i?
- Under the weaker linear model, what happens to $\mathbf{var}(\hat{y}_i \mid x_i)$?
- What happens to $\mathbf{var}(e_i \mid x_i)$?

INTUITION FOR LEVERAGE

Under the weaker linear model...

$$\begin{aligned}\mathrm{var}(e_i \mid x_i) &= (1 - h_{i,i})\sigma_\varepsilon^2 \\ \mathrm{var}(\hat{y}_i \mid x_i) &= h_{i,i}\sigma_\varepsilon^2 \\ \mathrm{cor}(\hat{y}_i, y_i \mid x_i) &= (h_{i,i})^{\frac{1}{2}}\end{aligned}$$

The larger $h_{i,i}$...

- The smaller the variability in the ith residual
- Less variability $\Rightarrow$ y_i tends to be closer to $\hat{y}_i$.
- The more $\hat{y}_i$ behaves like y_i.
- The closer $\hat{y}_i$'s variability is to that of y_i itself.

LEVERAGE AND "OUTLIERS IN X"

Through a bit of linear algebra (no need to derive), we can derive an alternative form for $h_{i,i}$ that helps understand what values of x_i have high leverage.

- Let H_0 be the "hat" matrix from a regression including only the intercept column.
- Let $X_{\setminus 0}$ be our design matrix **without** the intercept column (so it has p columns, one for each predictor, and n rows)
- Let $x_{i\setminus 0} = (x_{i1}, \ldots, x_{ip})$ be the predictor variables for the ith individual, with the intercept entry excluded
- Let $\bar{x}_{\setminus 0} = (\bar{x}_1, \ldots, \bar{x}_p)$ be the means of the predictor variables in our data

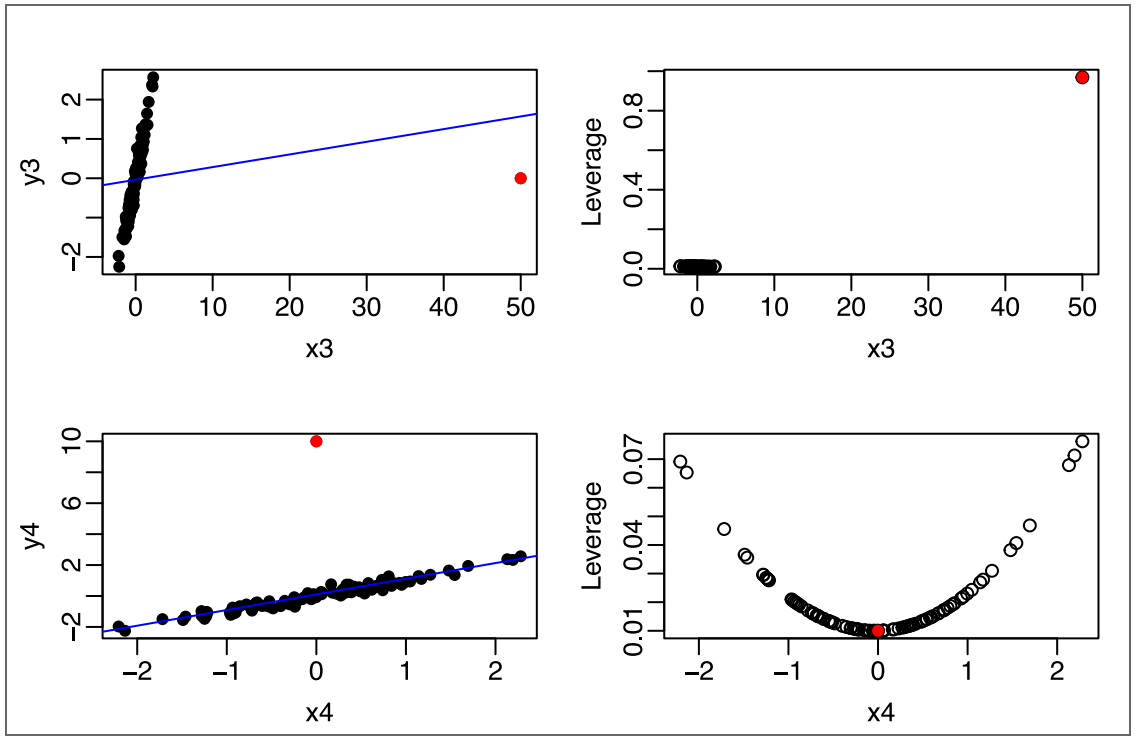
LEVERAGE AND "OUTLIERS IN X"

$$h_{i,i} = \frac{1}{n} + (x_{i\setminus 0} - \bar{x}_{\setminus 0})^\top (X_{\setminus 0}^\top (I - H_0) X_{\setminus 0})^{-1} (x_{i\setminus 0} - \bar{x}_{\setminus 0})$$

Second term is always non-negative

- If $x_i = \bar{x}$, second term cancels out, left with $\frac{1}{n}$.
- We see that $x_i = \bar{x}$ results in the smallest possible self-influence / leverage, $\frac{1}{n}$.
- As x_i moves farther away from $\bar{x}$, the leverage increases
- Points with high leverage are outliers with respect to the distribution of the **predictor variables**

LEVERAGE AND "OUTLIERS IN X"



USES FOR LEVERAGE

Leverages $h_{i,i}$ provide a scalar/numeric representation of the "self-influence" of a point

- How much does the observation y_i determine the fitted value $\hat{y}_i$?
  We see it is a function of the **predictor variables alone**, as it is based on the hat matrix.

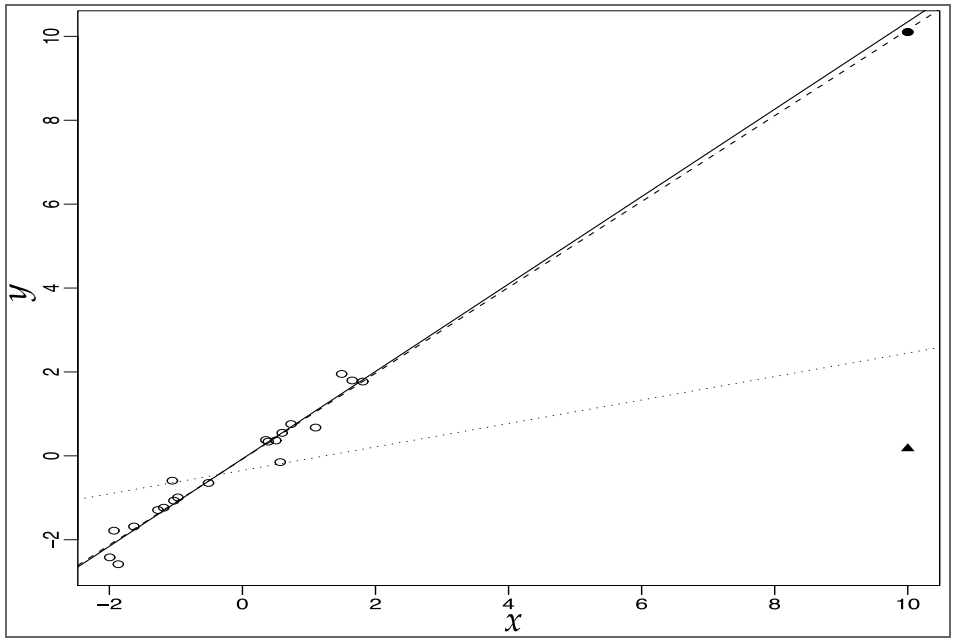
USES FOR LEVERAGE

Leverages help us answer the following

- "At what points x_i would an outlier in my responses y_i *have the potential to* dramatically alter my prediction equation?"
- Low leverage - outlying y_i unlikely to impact our prediction equation
- High leverage - outlying y_i can dramatically impact the prediction equation

Rule of thumb: Leverages greater than $\frac{2(p+1)}{n}$ are considered high.

WHICH POINT IS AN OUTLIER?



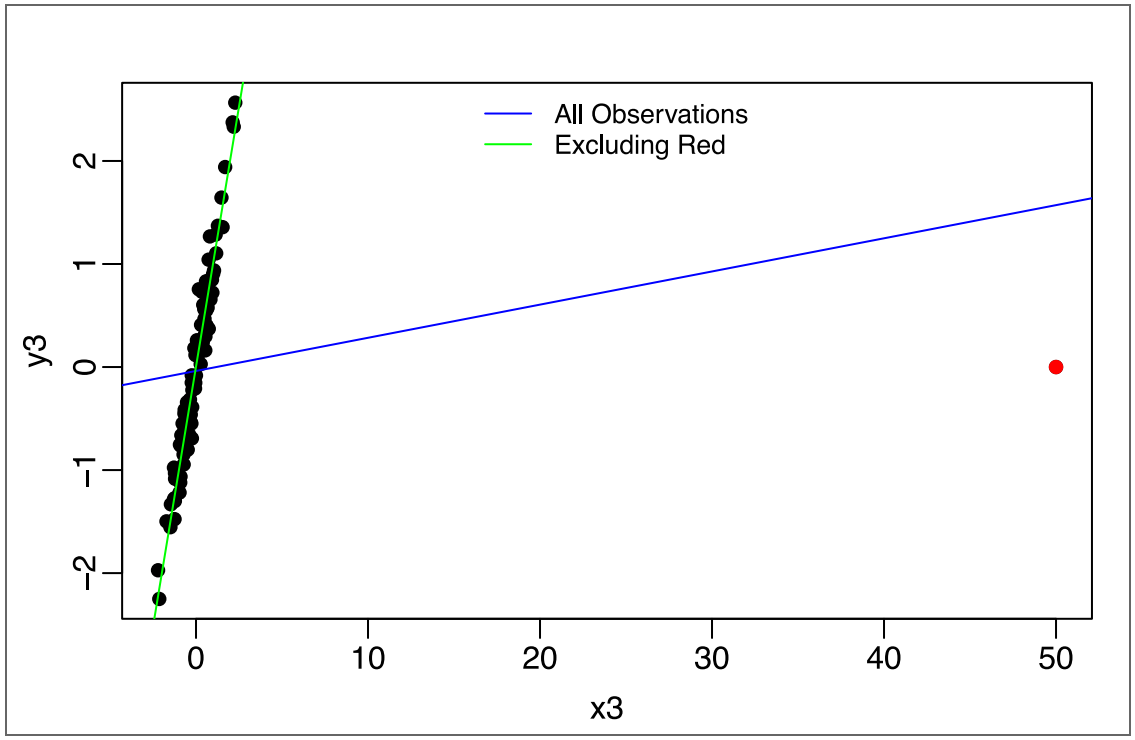
Can we come up with a numerical summary to help differentiate between these filled in points?

DETECTING OUTLIERS

How do we distinguish between truly unusual outcomes and large, but not unexpected, values for residuals?

- Issue: Outlying observations in the response may have small residuals if they also have high leverage.
- Idea: **Exclude point i**, and recompute $\hat{\beta}_{(i)}$ and $\hat{y}_{(i)} = x_i^\top \hat{\beta}_{(i)}$ with the ith observation excluded.
- Consequence: y_i is independent of $\hat{y}_{(i)}$ under the stronger linear model. (i wasn't used to form $\hat{y}_{(i)}$).
- If $|y_i - \hat{y}_{(i)}|$ is large, it suggests that the outcome y_i is an outlier...

REMOVING AN OBSERVATION



A TEST FOR OUTLIERS

Formally, we are testing

- **Null hypothesis**: For all observations $a = 1, \ldots, n, y_a = x_a^\top \beta + \varepsilon_a$, $\varepsilon_a \overset{\text{iid}}{\sim} N(0, \sigma_\varepsilon^2)$
- **Alternative hypothesis**: This model holds for all observations except for the ith individual, who comes from a different, unspecified, model.

A TEST FOR OUTLIERS

Under the null hypothesis, we have that

$$\mathrm{E}(y_i - \hat{y}_{(i)}) = 0$$
$$\mathrm{var}(y_i - \hat{y}_{(i)}) = \sigma_\varepsilon^2(1 + x_i^\top (X_{(i)}^\top X_{(i)})^{-1} x_i),$$

and that $y_i - \hat{y}_{(i)}$ is normally distributed.
[Does this remind you of the results underpinning our prediction intervals? It should...]

EXTERNALLY STUDENTIZED RESIDUALS

It turns out that, with $\hat{\sigma}_{\varepsilon(i)}$ the RMSE from a regression excluding the ith individual,

$$t_i = \frac{y_i - \hat{y}_{(i)}}{\hat{\sigma}_{\varepsilon(i)} \left\{ 1 + x_i^\top (X_{(i)}^\top X_{(i)})^{-1} x_i \right\}^{\frac{1}{2}}}$$
$$\sim t_{n-(p+1)-1}$$

EXTERNALLY STUDENTIZED RESIDUALS

These t_i are called **externally studentized** residuals. Interestingly, t_i can also be expressed as

$$t_i = r_i \left(\frac{n - (p+1) - 1}{n - (p+1) - r_i^2} \right)^{1/2},$$

where r_i are the standardized residuals we introduced previously.

- Implication: we can actually compute t_i **without** computing a regression that excludes the ith individual.

MULTIPLE HYPOTHESIS TESTS

For any particular i,

- We could then compute $|t_i|$ and compare it to $t_{1-\alpha/2,(n-p-2)}$.
- Reject if $|t_i|$ exceeds this value
- For any particular i, this controls the Type I error rate (the probability of falsely rejecting a true null hypothesis) at α.

MULTIPLE HYPOTHESIS TESTS

Suppose I repeated this process for all n observations in my study. For each observation, I try to determine whether that individual observation is an outlier.

- What do you think the probability of committing at least one Type I error will be?
- α? Larger than α?

BONFERRONI CORRECTION

Suppose that the stronger linear model actually holds for all n observations, meaning that none of the points are actually generated from a different model.

$$\begin{aligned}\text{Type I error rate} &= \mathbf{P}(\text{reject at least one test} \mid H_0 \text{ True}) \\ &\leq \textstyle\sum_i \mathbf{P}(\text{reject test } i \mid H_0 \text{ True}) \\ &= n\alpha\end{aligned}$$

Bonferroni correction: test each hypothesis at level α/n

- Applying the above result: if each test is conducted at α/n, the probability of committing at least one Type I error is bounded above by α.

WHAT TO DO WITH OUTLIERS?

In general, it is **not** good practice to remove outliers unless we have a compelling reason that the point isn't representative of the population of interest

- Good reason: data entry error, individual was included in data set who truly should have been excluded

WHAT TO DO WITH OUTLIERS?

In general, it is **not** good practice to remove outliers unless we have a compelling reason that the point isn't representative of the population of interest

- Bad reasons
- Slope coefficients become more significant when excluded
- Data looks more linear when observation excluded
- R^2 increases when observation excluded
- ...

Outliers can indicate that your linear model doesn't apply to the entirety of the population under study, that there might be a lurking variable, that you might need a more complex model...

COMBINING LEVERAGE AND OUTLIERS

For an observation to be truly **influential** in changing altering the fitted regression equation, it needs to be at least one of the following

- Outlier
- Large leverage point

We've also seen that

- Some outliers don't affect the regression equation particularly much (if they are low leverage points)
- Some large leverage points don't end up impacting the fitted model too much (if their outcome follows the trend in the rest of the data)

Can we devise a measure to detect such influential points?

COOK'S DISTANCE

The Cook's distance D_i of an observation i is the sum of the squared differences between predicted values in regressions fit including i and excluding i, normalized by $(p+1)\hat{\sigma}_\varepsilon^2$

$$D_i = \frac{(X\hat{\beta} - X\hat{\beta}_{(i)})^\top (X\hat{\beta} - X\hat{\beta}_{(i)})}{(p+1)\hat{\sigma}_\varepsilon^2},$$

where $\hat{\beta}_{(i)}$ are the regression coefficients from a regression excluding observation i.

COOK'S DISTANCE

Equivalently, Cook's distance can be expressed as

$$D_i = \frac{1}{p+1} r_i^2 \frac{h_{i,i}}{1-h_{i,i}},$$

from whence we see that D_i combines the impact of

- Large residual (in terms of standardized residual r_i)
- Large leverage ($h_{i,i}$)

Rule of thumb: $D_i > 1$ indicates an influential point

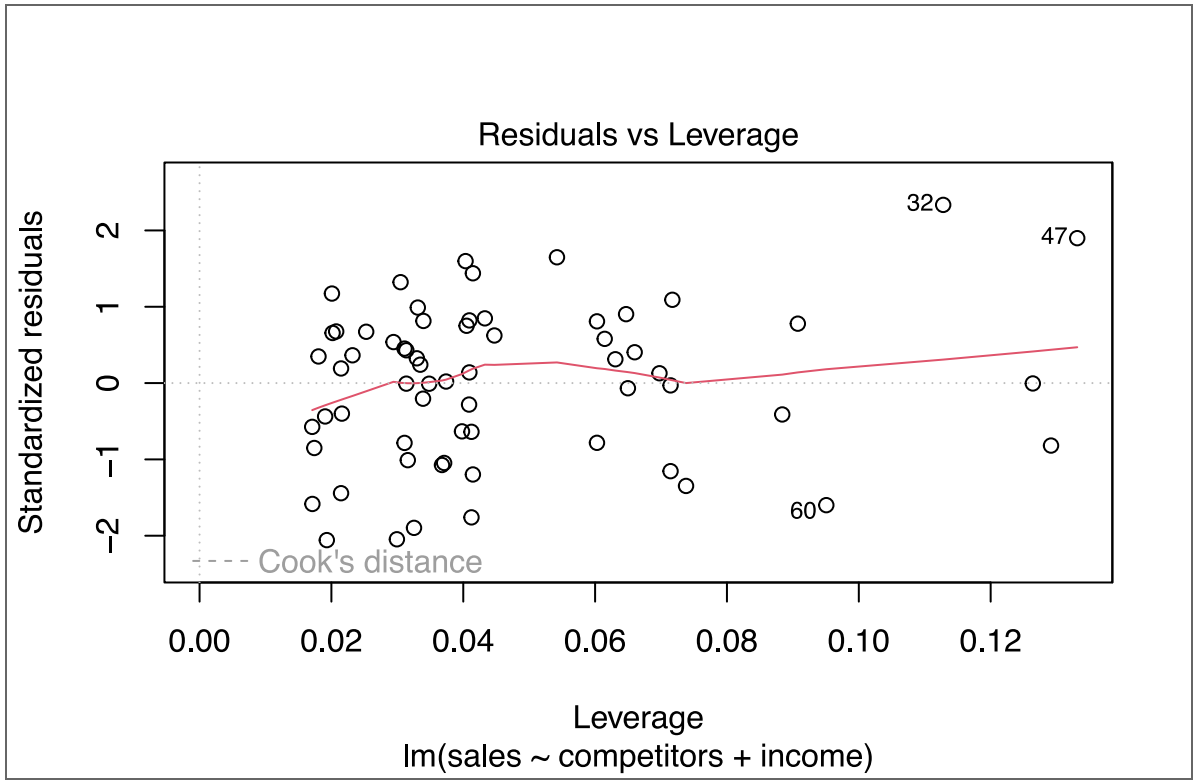
A DIAGNOSTIC PLOT FOR INFLUENTIAL POINTS

A Residuals vs Leverage plot visualizes

- Standardized residuals r_i (y axis)
- Value for Leverage (x axis)
- Cook's Distance (contours for $D_i = 0.5$ and $D_i = 1$).

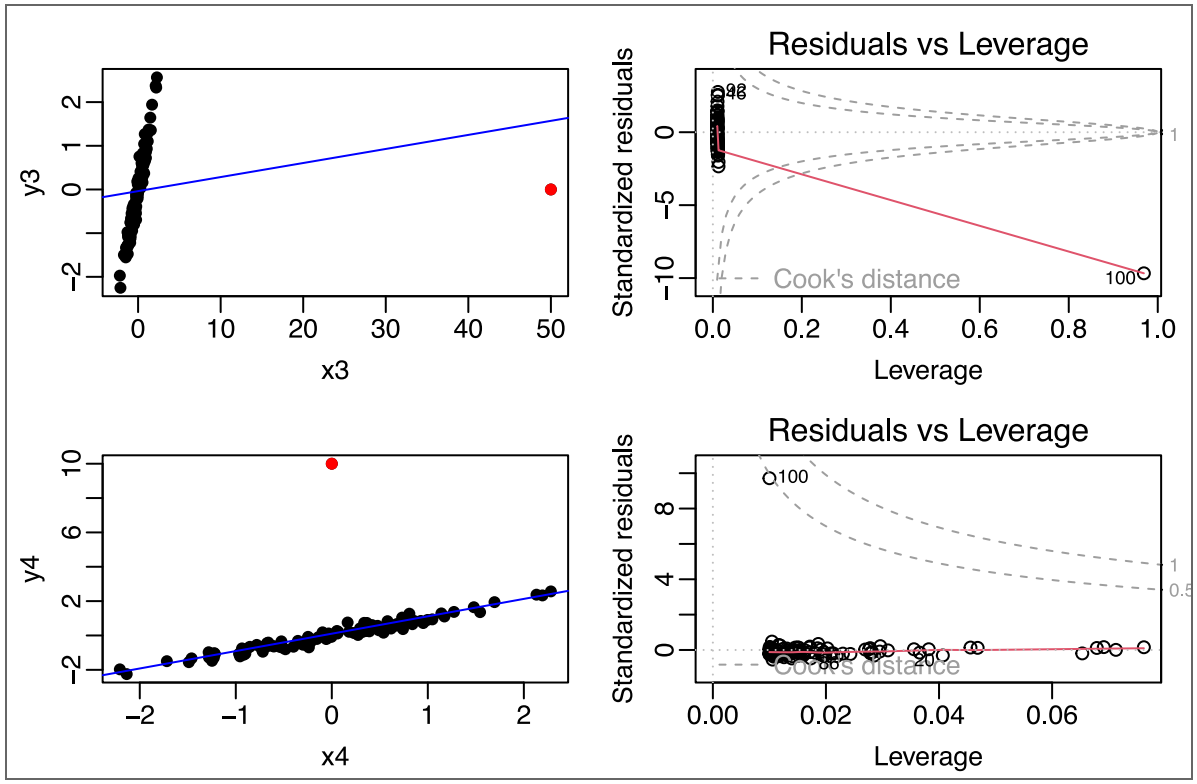
RESIDUALS VS LEVERAGE

Here's a normal looking residuals vs leverage plot. Nothing concerning here...



RESIDUALS VS LEVERAGE

In our simulated data sets:



Speaker notes